

Thm. Riemann's Theorem.

Let f be a complex function having an isolated singularity at $z=z_0$.

If f is bounded for some deleted neighborhood of z_0 ,

then z_0 is removable.

Proof. By the hypothesis, for some $R > 0$, f is analytic and bounded in

$$0 < |z - z_0| < R.$$

Given $z_1 \in B_R(z_0) \setminus \{z_0\}$, let $r > 0$ s.t. $r < |z_1 - z_0| < R$. So, f is analytic and bounded in the annulus between $C_1: |z - z_0| = R$ and $C_2: |z - z_0| = r$.

By the Cauchy's integral formula,

$$f(z_1) = \frac{1}{2\pi i} \cdot \left[\int_{C_1} \frac{f(u)}{u - z_1} du - \int_{C_2} \frac{f(u)}{u - z_1} du \right]$$

Meanwhile, by the ML inequality,

$$\left| \int_{C_2} \frac{f(u)}{u - z_1} du \right| \leq \sup |f| \cdot 2\pi r \cdot \frac{1}{|z_1 - z_0| - r} \rightarrow 0 \text{ as } r \rightarrow 0.$$

$$\text{So, } f(z_1) = \frac{1}{2\pi i} \int_{C_1} \frac{f(u)}{u - z_1} du.$$

Since z_1 was arbitrary, we can denote that

$$f(z) = \frac{1}{2\pi i} \cdot \int_{C_1} \frac{f(u)}{u - z} du \quad \text{for all } 0 < |z - z_0| < R,$$

$$\text{So, define } f(z_0) = \frac{1}{2\pi i} \cdot \int_{C_1} \frac{f(u)}{u - z_0} du, \quad \text{we obtain the result.}$$

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Thm. If f has an isolated singularity at z_0 and $f(z) \rightarrow \infty$ as $z \rightarrow z_0$, then $f(z)$ has a pole at $z = z_0$.

Proof. By the assumption, given $R > 0$, there is $\delta > 0$ such that

$$0 < |z - z_0| < \delta \Rightarrow |f(z)| > R.$$

put $g(z) = \frac{1}{f(z)} \quad (0 < |z - z_0| < \delta).$

Then g is analytic and bounded by $\frac{1}{R}$. So, z_0 is removable for g ,

$$g(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \quad |z - z_0| < \delta$$

and $a_0 = 0$, since $\lim_{z \rightarrow z_0} g(z) = \lim_{z \rightarrow z_0} 1/f(z) = 0$.

And, since $g \neq 0$ for $0 < |z - z_0| < \delta$, not all the coefficients of g are zero.

So, for some $k \geq 1$, $a_k \neq 0$ with

$$g(z) = a_k \cdot (z - z_0)^k + a_{k+1} (z - z_0)^{k+1} + \dots$$

so that

$$\frac{1}{(z - z_0)^k \cdot f(z)} = a_k + a_{k+1} (z - z_0) + \dots \rightarrow a_k \quad \text{as } z \rightarrow z_0.$$

Since a_k is non zero, $(z - z_0)^k \cdot f(z) \rightarrow a_k^{-1}$. $\square$

Corollary. If f has a pole at $z = z_0$ order m , (i.e., $\lim_{z \rightarrow z_0} (z - z_0)^m \cdot f(z)$ exists in $\mathbb{C} \setminus \{0\}$.)

Then, $f(z) = \sum_{n=-m}^{\infty} b_n \cdot (z - z_0)^n$.

Isolated singularity at ∞ .

If f is analytic for all $R < |z| < \infty$, for some $R > 0$, then we say that f has an isolated singularity at ∞ .

For this $R > 0$, observe that

$f(z)$ is analytic at $|z| > R$ if and only if $f(\frac{1}{z})$ is analytic at $|z| < \frac{1}{R}$.

$f(z)$ has an isolated singularity at ∞ iff $f(\frac{1}{z})$ has an isolated singularity at 0.

Thm. Let f be a non-constant entire function. Then,

f is polynomial if and only if $f(z) \rightarrow \infty$ as $z \rightarrow \infty$.

Proof. Suppose that f is a polynomial. WLOG, let $f(z) = a_0 + a_1 z + \dots + a_n z^n$. ($a_n \neq 0$).

Then, $f(\frac{1}{z}) = \frac{a_n}{z^n} + \frac{a_{n-1}}{z^{n-1}} + \dots + \frac{a_1}{z} + a_0$ so that $\lim_{z \rightarrow 0} z^n \cdot f(\frac{1}{z}) = a_n \neq 0$.

So, $f(z)$ has a pole at ∞ , so $\lim_{z \rightarrow \infty} f(z) = \infty$.

Conversely, suppose that: $f(\frac{1}{z})$ has a pole of order m at 0. That is,

$$\lim_{z \rightarrow 0} \frac{f(z)}{z^m} = A \in \mathbb{C} \setminus \{0\}.$$

That is, for some $R > 0$, $|z| \geq R \Rightarrow \left| \frac{f(z)}{z^m} - A \right| < 1$

$$\Rightarrow \left| \frac{f(z)}{z^m} \right| < 1 + |A|$$

$$\Rightarrow |f(z)| < (1 + |A|) \cdot z^m$$

So, f is a polynomial of degree at most m .

